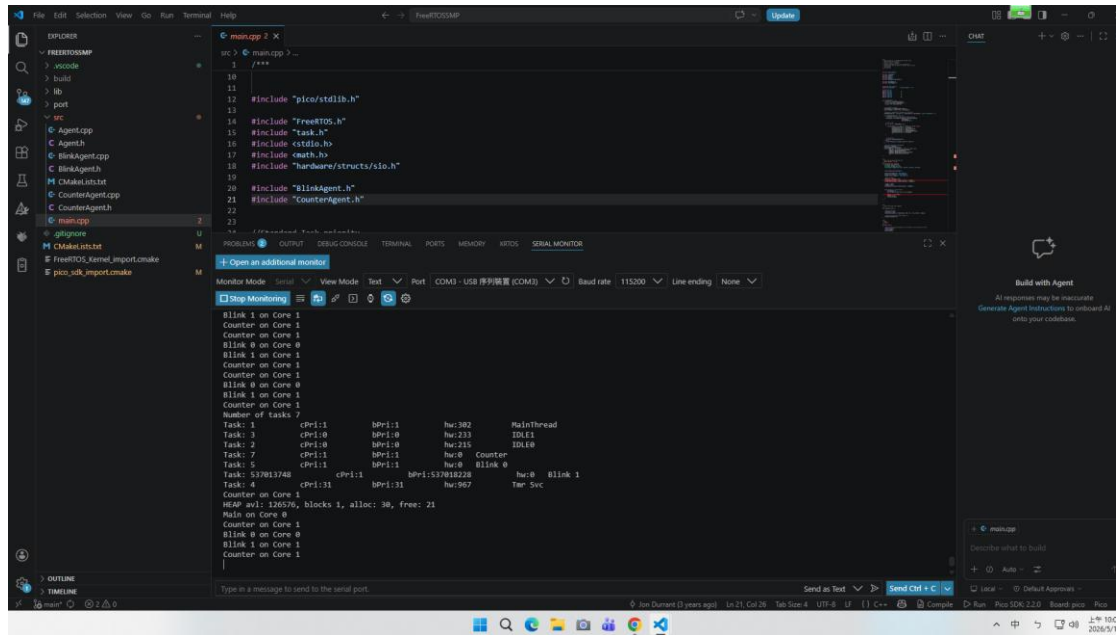


OS-8

U13157004



The screenshot shows the FreeRTOS IDE interface. The main window displays the `main.cpp` file, which includes the following code:

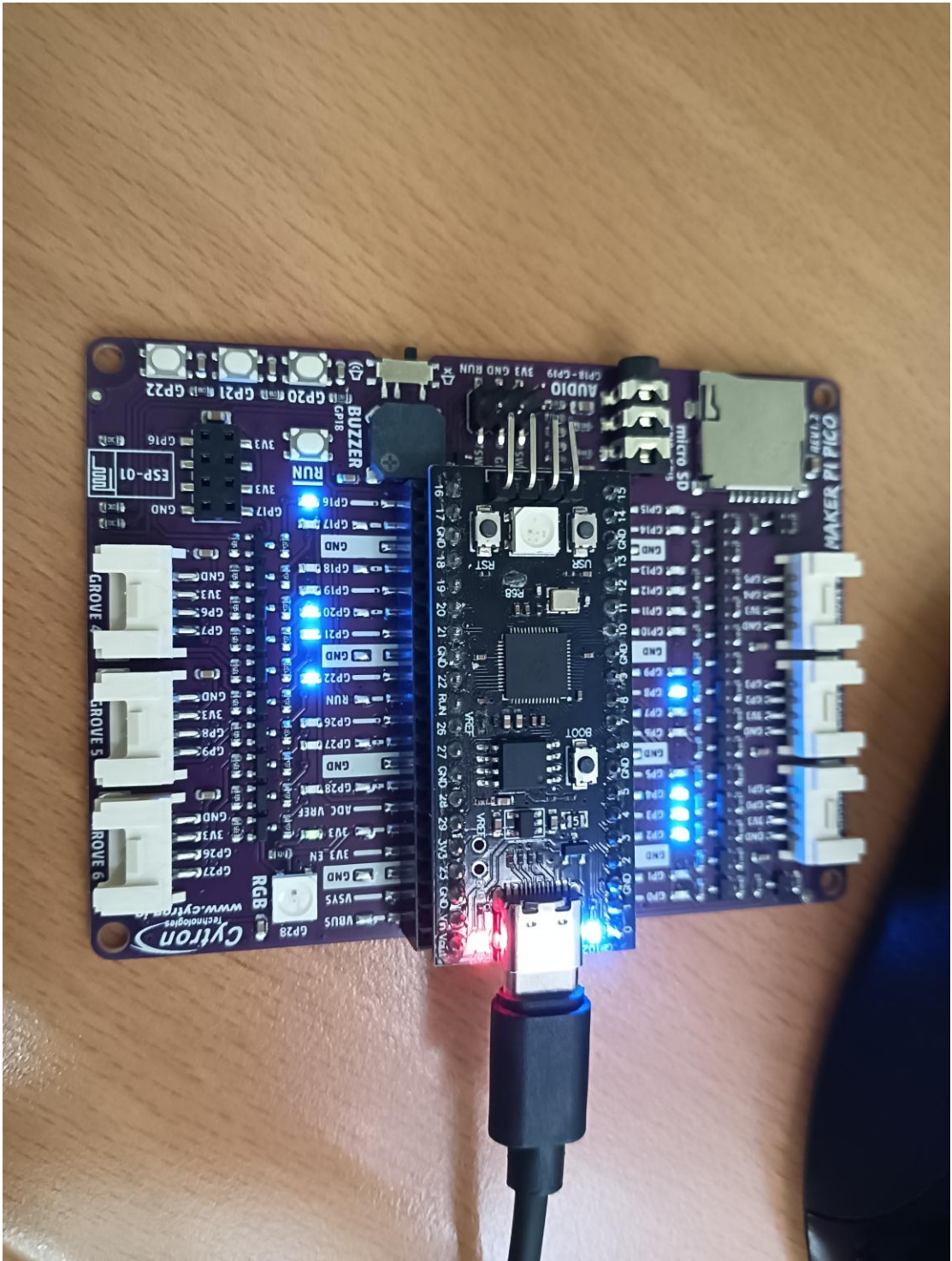
```
1  /**
2  *
3  *
4  *
5  *
6  *
7  *
8  *
9  *
10 *
11 *
12 *include "pico/stdlib.h"
13 *
14 *include "FreeRTOS.h"
15 *include "task.h"
16 *include <stdio.h>
17 *include <math.h>
18 *include "hardware/structs/sio.h"
19 *
20 *include "BlinkAgent.h"
21 *include "CounterAgent.h"
22 *
23 */
24
25 // Threaded tasks
26
27 // Open an additional monitor
28
29 // Stop Monitoring
30
31 Blink 1 on Core 1
32 Counter on Core 1
33 Counter on Core 1
34 Blink 0 on Core 0
35 Blink 1 on Core 1
36 Counter on Core 1
37 Counter on Core 1
38 Blink 0 on Core 0
39 Blink 1 on Core 1
40 Counter on Core 1
41
42 Number of tasks: 7
43 Task: 1 cPri:1 bPri:1 hwi:302 MainThread
44 Task: 3 cPri:0 bPri:0 hwi:233 IDLE1
45 Task: 2 cPri:0 bPri:0 hwi:235 IDLE0
46 Task: 7 cPri:1 bPri:1 hwi:0 Counter
47 Task: 5 cPri:1 bPri:1 hwi:53784228 hwi:0 Blink 0
48 Task: 4 cPri:1 bPri:1 hwi:53784228 hwi:0 Blink 1
49 Task: 6 cPri:1 bPri:1 hwi:967 Task Svc
50
51 Counter on Core 1
52 Heap avl: 126576, blocks 1, alloc: 30, free: 21
53 Main on Core 0
54 Counter on Core 1
55 Blink 0 on Core 0
56 Blink 1 on Core 1
57 Counter on Core 1
```

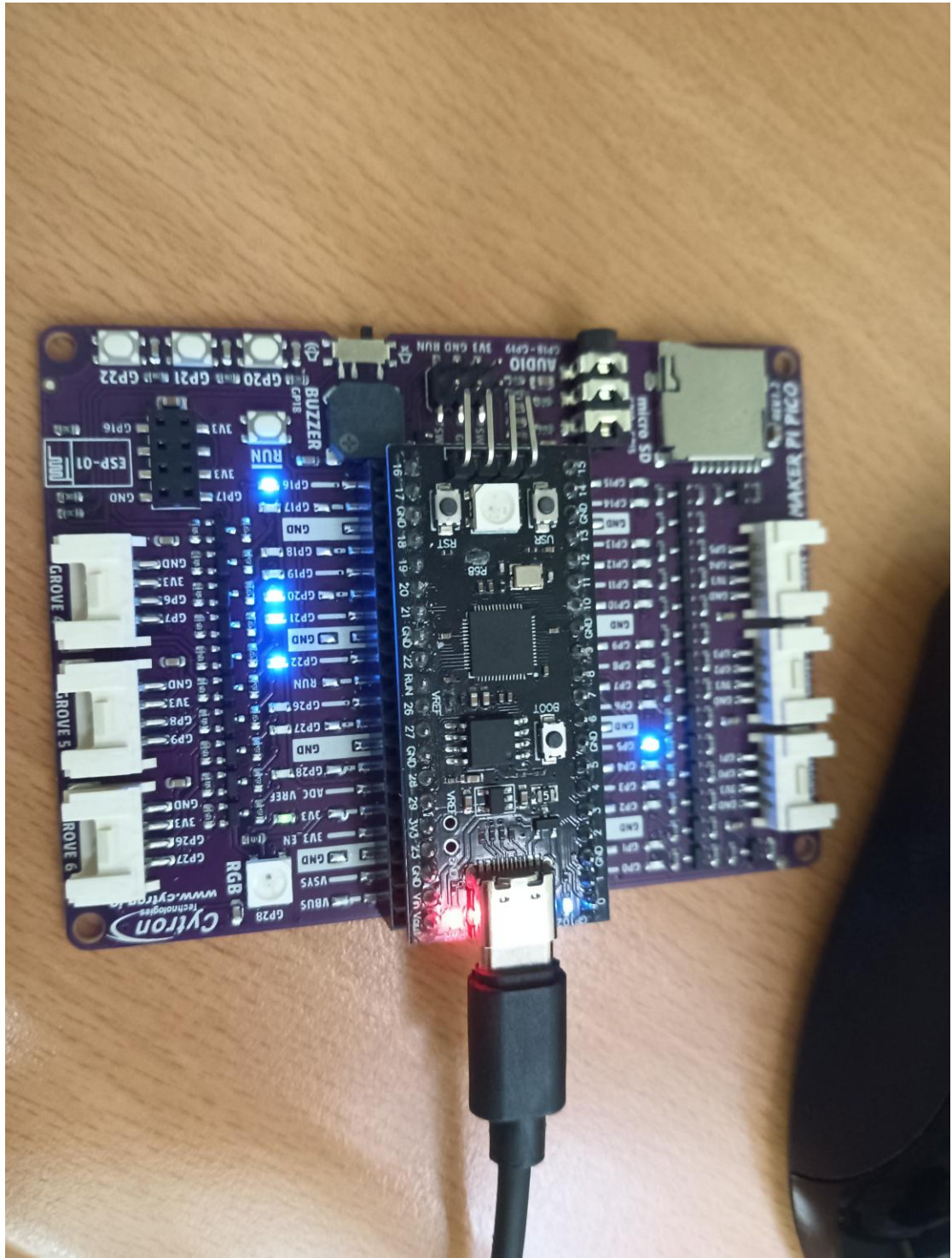
The Serial Monitor window shows the output of the program, which includes the task list and the status of the tasks. The output is as follows:

```
Blink 1 on Core 1
Counter on Core 1
Counter on Core 1
Blink 0 on Core 0
Blink 1 on Core 1
Counter on Core 1
Counter on Core 1
Blink 0 on Core 0
Blink 1 on Core 1
Counter on Core 1

Number of tasks: 7
Task: 1 cPri:1 bPri:1 hwi:302 MainThread
Task: 3 cPri:0 bPri:0 hwi:233 IDLE1
Task: 2 cPri:0 bPri:0 hwi:235 IDLE0
Task: 7 cPri:1 bPri:1 hwi:0 Counter
Task: 5 cPri:1 bPri:1 hwi:53784228 hwi:0 Blink 0
Task: 4 cPri:1 bPri:1 hwi:53784228 hwi:0 Blink 1
Task: 6 cPri:1 bPri:1 hwi:967 Task Svc

Counter on Core 1
Heap avl: 126576, blocks 1, alloc: 30, free: 21
Main on Core 0
Counter on Core 1
Blink 0 on Core 0
Blink 1 on Core 1
Counter on Core 1
```





README

這段代碼是 Raspberry Pi Pico (RP2040) 使用 **FreeRTOS SMP (對稱多處理)** 的典型範例。它展示了如何將不同的任務 (Tasks) 分配到 Pico 的兩個核心 (Core 0 和 Core 1) 上執行。

- ****雙核平行運算****: 利用 FreeRTOS SMP 技術，將閃爍任務與計數任務分散至 Core 0 與 Core 1。
- ****實時監測****: 整合 `runTimeStats` 功能，透過序列埠實時追蹤系統堆疊安全與任務調度。
- ****隨機化邏輯****: Counter 任務結合 `rand()` 函數，動態呈現 LED 狀態切換。